



GIRIJANANDACHOWDHURYUNIVERSITY

Hathkhowapara, Azara, Guwahati 781017, Assam

Department of Civil Engineering

Experiment -1

BCE235PE10P

OBJECTIVE : To attend at least two demonstrations by reputable manufacturers of water supply pipes and fittings, and to perform cutting and jointing of pipes as per recommended procedures and given dimensions

EQUIPMENT AND MATERIALS REQUIRED :

- Water supply pipes (PVC, GI, or HDPE types as specified)
- Fittings (elbows, tees, unions, couplings, nipples)
- Pipe cutter, hacksaw, and pipe threading tool
- Reaming tool, round file, and deburring tool
- Measuring tape and marker
- Teflon (PTFE) tape and jointing compound (for GI pipes)
- Cleaning pads or brushes
- Solvent cement (for PVC) or butt fusion tools (for HDPE)
- Pipe wrench and pipe stand
- Safety gear: gloves, goggles, and helmet ♦♦♦.

Demonstration : Attend demonstrations from at least two reputed manufacturers (for example, Finolex, Astral, Supreme, Jain Irrigation, or APL Apollo).

OBSERVE :

- Different types of pipe materials and their uses.
- Manufacturer-recommended jointing procedures (solvent welding, threaded joints, or fusion welding).
- Quality checks after jointing, such as leak tests and pressure ratings

PROCEDURE :

- **Step 1: Measuring** Measure pipe length as per given dimensions using measuring tape. Mark the cutting point clearly using a permanent marker or scribe ♦
- **Step 2: Cutting** Use a disc-type pipe cutter or hacksaw for cutting pipes square to the axis. For smooth operation, apply cutting oil to metal pipes. Avoid excessive pressure to prevent deformation ♦♦♦.
- **Step 3: Reaming and Deburring** Smoothen inner and outer edges using a reamer or round file. Remove burrs carefully to avoid restricting flow or damaging fittings ♦.
- **Step 4: Cleaning** Clean pipe ends and fittings with sand cloth or abrasive pad. Ensure joints are free from dust, grease, or oil ♦.
- **Step 5: Jointing** For PVC pipes: Use solvent cement suitable for the pipe grade. Apply evenly on both spigot and socket surfaces, push and hold firmly for a few seconds ♦. For GI pipes: Apply Teflon tape or white lead with jute thread on male threads, then tighten into socket or fitting using a wrench ♦. For HDPE pipes: Use butt-fusion welding. Align ends, heat to fusion temperature (~220°C), press together, and allow to cool ♦. Ensure alignment and insertion depth are as per specifications.
- **Step 6: Testing** After jointing, conduct a pressure test or water leak test to ensure joint integrity



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OBSERVATION & CONCLUSION:

- Record methods used by each manufacturer and compare joint quality.
 - Note differences in techniques between PVC, GI, and HDPE systems.
 - Conclude which jointing method produced leak-free and durable results.
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Experiment -2

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OBJECTIVE : To measure and compare the water flow rates of various plumbing fixtures and fittings installed in residential, commercial, and institutional toilets, and to analyze their conformity to standard performance limits

EQUIPMENT AND MATERIALS REQUIRED :

- Flow measuring container or flow meter (marked in liters)
- Stopwatch (with least count of 0.01 s)
- Measuring cylinder (1-10 L capacity)
- Data recording sheet
- Access to residential, commercial, and institutional toilets
- Calculator and conversion chart
- Pressure gauge for in-line systems
- Flow restrictor or aerator (for comparison) .

THEORY :

The flow rate (Q) is the volume of water (V) discharged per unit time (t):

$$Q = V/t$$

where: [Q] = flow rate (L/min),

[V] = volume of water collected (L),

[t] = time taken (minutes) .

PROCEDURE :

- Identify types of fixtures (faucets, showers, urinals, etc.) in each facility. Note the manufacturer, type (sensor/manual), and installation type.
- Measurement of Flow Rate Place a measuring container under the outlet. Open the fixture fully and simultaneously start the stopwatch. Collect water for exactly 10 seconds and then close the fixture. Measure the volume of water collected. Convert to liters per minute (L/min) using $[Q = V / t \times 60]$.
- Measurement of Flush Volume Empty the toilet tank. Fill the tank and record the total volume using a measuring scale or tank markings. Flush once completely and record water used per flush. Repeat for at least three readings and find the average .
- Repeat the above for each fixture in residential, commercial, and institutional buildings.



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OBSERVATION :

Observations

Prepare a table as follows:

Fixture Type	Location	Flow/Flush Measured (L/min)
Wash Basin Tap	Residential	7.5 L/min
Shower	Commercial	6 L/min
WC (Dual Flush)	Institutional	4.8 L/flush

CALCULATION :

For example,

If volume collected = 1.2 L in 10 seconds:

$$Q = 1.2 \div 10 \times 100$$

$$= 7.2 \text{ L/min.}$$

If flush tank discharges 5.8 L in one complete flush,

then flush volume = 5.8 L/flush

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Experiment -3

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OBJECTIVE : To understand the manufacturer-recommended procedures for the installation and jointing of DWV pipes, traps, and fittings, and to perform cutting and jointing operations for the given pipe dimensions following standard plumbing practices

EQUIPMENT AND MATERIALS REQUIRED :

- DWV pipes (PVC-U, ABS, or cast Iron types)
- Fittings (traps, elbows, tees, reducers, cleanouts, adapters)
- Pipe cutter or fine-tooth hacksaw
- Deburring tool, file, and beveling tool
- Cleaning solvent and lint-free cloth
- Solvent cement or rubber ring joint lubricant (depending on pipe type)
- Measuring tape, marker, and depth gauge
- Safety goggles, gloves, and apron
- Rubber ring (for RRJ method)
- Pipe stand and alignment guides

DEMONSTRATION:

- Attend demonstration by a reputed manufacturer such as Marley, Iplex, Charlotte Pipe, Spears, or IPEX.
- Observe and record: Types of DWV pipe materials used (PVC-U, ABS, cast iron, etc.) Recommended jointing techniques (solvent weld or rubber ring joint) Inspection and testing procedures for drainage installations.
- Learn about below-ground and above-ground systems and how ventilation and traps prevent sewer gases from entering habitable areas.
- The flow rate (Q) is the volume of water (V) discharged per unit time (t):

$$Q = V/t$$

where: [Q] = flow rate (L/min),

[V] = volume of water collected (L),

[t] = time taken (minutes).

PROCEDURE :

Step 1: Measurement

- Measure the required length of pipe using a measuring tape.
- Mark the cut line perpendicular to the pipe's axis.

Step 2: Cutting

- Cut the pipe square using a pipe cutter or hacksaw.
- Avoid uneven cuts which may cause poor joint sealing ♦.

Step 3: Deburring and Beveling

- Remove burrs from both inner and outer edges. Bevel (chamfer) the edge at a 10°–15° angle for easy insertion into the fitting ♦♦.

Step 4: Cleaning

- Wipe the pipe ends and fitting sockets with a clean, dry cloth.
- Remove any dirt, oil, or moisture.

Step 5: Dry Fit

- Check alignment by inserting the pipe spigot into the fitting socket (should insert about 1/3 to 2/3 depth).
- Mark the insertion line on the pipe (witness mark) for reference



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Step 6: Jointing

- Method A – Solvent Welded Joint (for PVC or ABS DWV) Apply primer (for PVC only) evenly on pipe and fitting sockets. Spread solvent cement uniformly over both surfaces. Quickly insert the pipe into the fitting with a quarter turn to distribute cement. Hold in position for a few seconds to prevent pushout. Wipe off excess cement. Allow curing as per cement manufacturer's specifications (usually 15–30 min) ♦♦.
- Method B – Rubber Ring Joint (RRJ, for larger or underground DWV pipes) Clean rubber rings and sockets thoroughly. Apply lubricant evenly to spigot end up to the witness mark. Insert the pipe into the socket until the mark is just visible. Ensure allowance for expansion and contraction ♦♦

Step 7: Trap Assembly

- Select suitable traps (P-trap, S-trap, or bottle trap) and connect between fixture and drainage line.
- Ensure water seal depth (typically 50 mm) is maintained to prevent gas leakage.

Step 8: Alignment and Support

- Ensure proper slope of drainage pipes (1:40 to 1:60 gradient).
- Provide adequate support spacing based on pipe size and material ♦♦.

Step 9: Testing

- Perform a leak test or hydrostatic test after joints have cured.
- Visually inspect all connections for alignment, seal integrity, and absence of cracks.

Observations and Report :

- Record pipe type, size, and jointing method used.
- Note curing time, fit quality, and observed test results.
- Compare performance between solvent weld and rubber ring joints.
- Document any defects or misalignment issues.

Conclusion :

- Solvent welded joints are ideal for small-diameter, above-ground DWV systems, offering permanent airtight seals.
- Rubber ring joints suit large-diameter and underground pipes, allowing thermal expansion movement.
- Proper cutting, beveling, cleaning, and alignment are critical for leak-free, durable installations.

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Experiment -4

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OBJECTIVE : To perform a site visit to a plumbing installation project and document detailed observations on the plumbing system, architectural and structural provisions, pipe materials, work methods, site conditions, safety measures, and to provide recommendations in compliance with UIPC-I and NBC standards.

EQUIPMENT AND MATERIALS REQUIRED :

- Site visit checklist structured as per UIPC-I/NBC standards
- Measuring tape
- Camera or mobile phone for photographs
- Notebook or digital device for notes
- Personal protective equipment (PPE) for site safety (helmet, gloves, boots, goggles)

PROCEDURE :

1. Introduction and Preparation

- Identify the site location and nature of plumbing installation (residential, commercial, institutional).
- Review architectural drawings, plumbing plans, and specifications beforehand to understand scope and design intent.

2. Inspect Plumbing System Components

- Check water supply system: materials used (PPR, CPVC, GI, PVC, etc.), pipe diameters, routing, insulation.
- Inspect drainage and waste systems: pipe types (PVC-U, HDPE), venting arrangements, slope and gradient compliance.
- Verify installation of fixtures and their compliance with flow/flush standards.
- Check valves, fittings, joints for leakage or defects. Observe backflow prevention measures, pressure regulating devices.

3. Architectural and Structural Provisions

- Verify embedding/protection of pipes in walls, floors or slabs as per standards. Check pipe supports, hangers, and clamps—spacing and quality. Confirm provision for access panels, inspection chambers, and cleanouts.
- Review alignment, avoiding horizontal runs without proper slope.

4. Pipe Materials and Work Methods

- Note types and quality of pipes and fittings used.
- Observe cutting, jointing, and sealing methods (solvent welding, threaded joints, rubber ring joints).
- Verify curing times for solvent joints and appropriate tools usage.
- Confirm adherence to manufacturer guidelines and UIPC/NBC specifications.

5. Site Conditions and Safety

- Check housekeeping and cleanliness of the work area.
- Inspect storage and handling of pipes/materials onsite.
- Observe safety precautions for workers: PPE usage, barricades, warning signs.
- Identify hazards such as wet floors, electrical risks near plumbing zones.



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BCE235PE10P

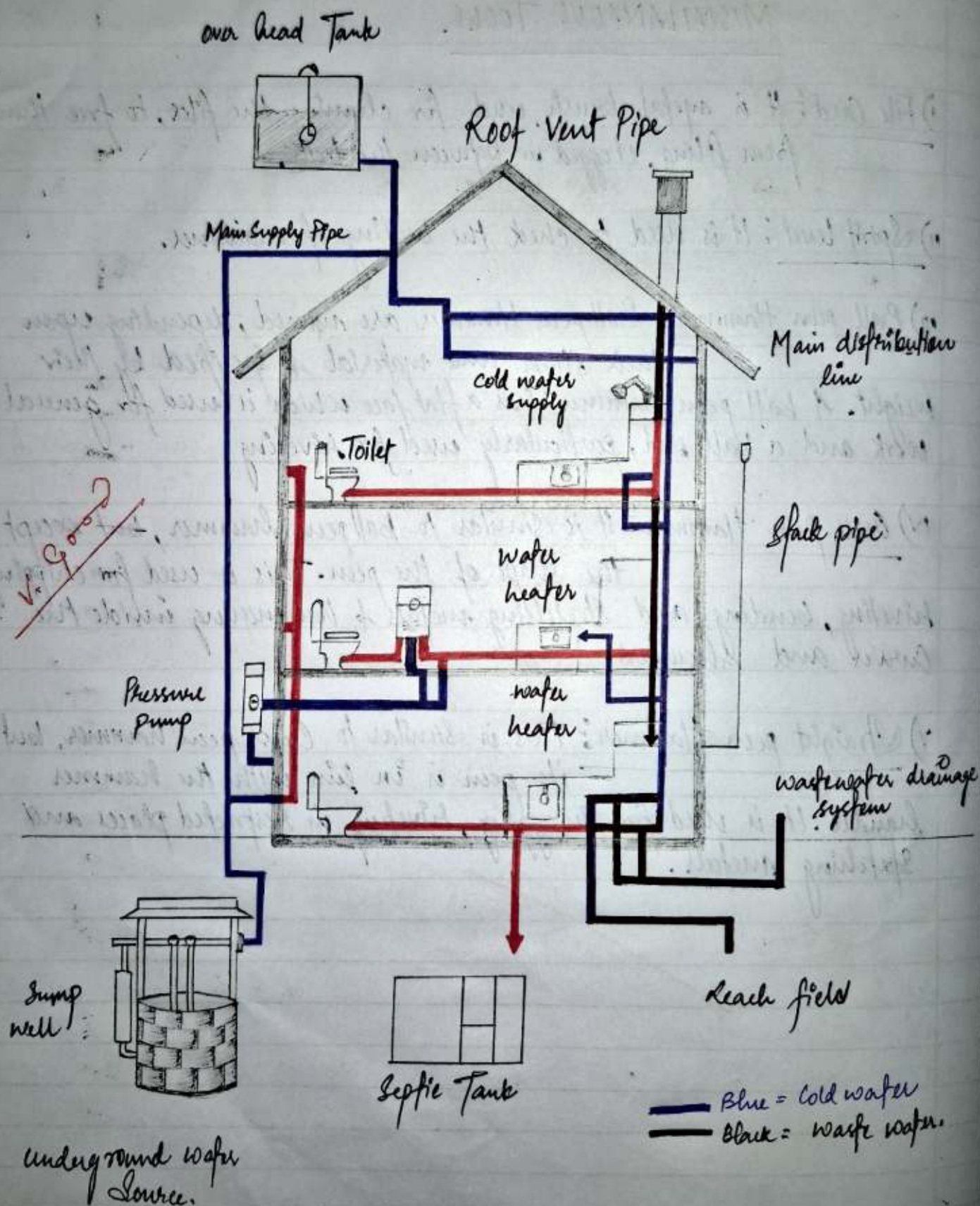
6. Testing and Quality Assurance

- Observe hydrostatic or air pressure testing for pipes post-installation. Note test pressures and durations as per code requirements.
- Check for leakages or joint failures during tests.

7. Recommendations

- Suggest improvements in pipe routing or supports if non-compliant. Recommend enhanced safety measures if gaps observed.
- Advise verifying material certifications and proper testing documentation.
- Ensure all practices are in line with UIPC-I/NBC plumbing and safety norms.

Safety Precautions During Visit: Always wear required PPE. Maintain safe distance from heavy machinery or active construction zones. Follow site-specific safety protocols.



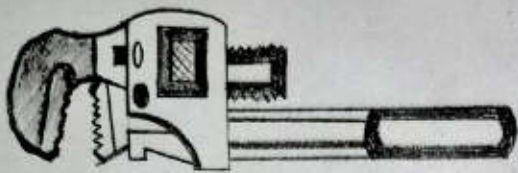


Fig: Pipe Wrench

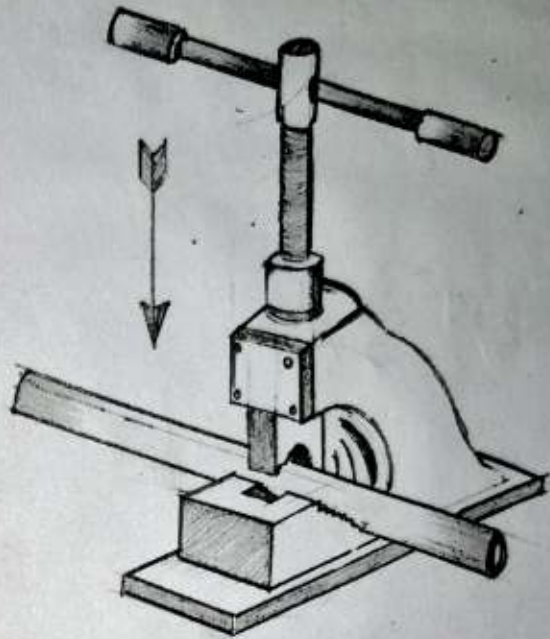


Fig: Pipe Vice

✓ Good

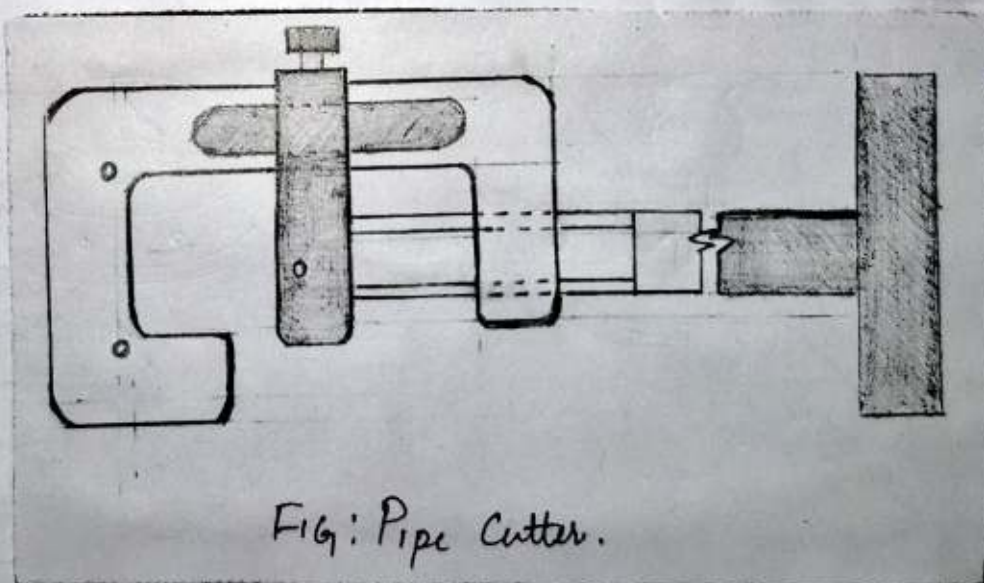
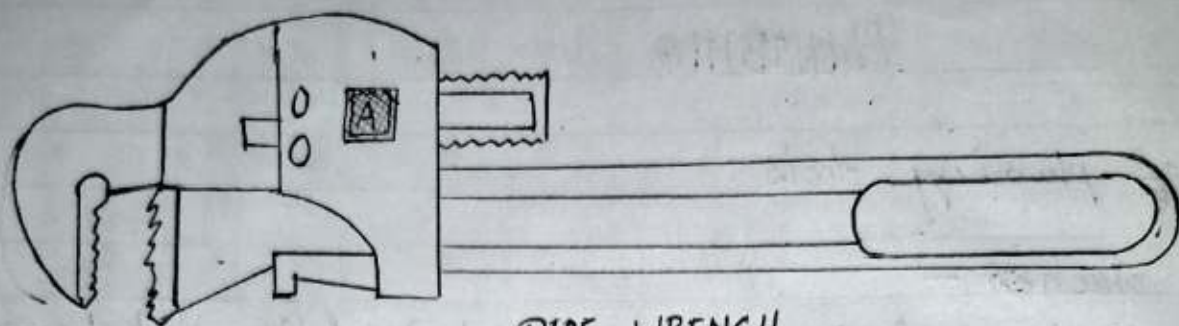
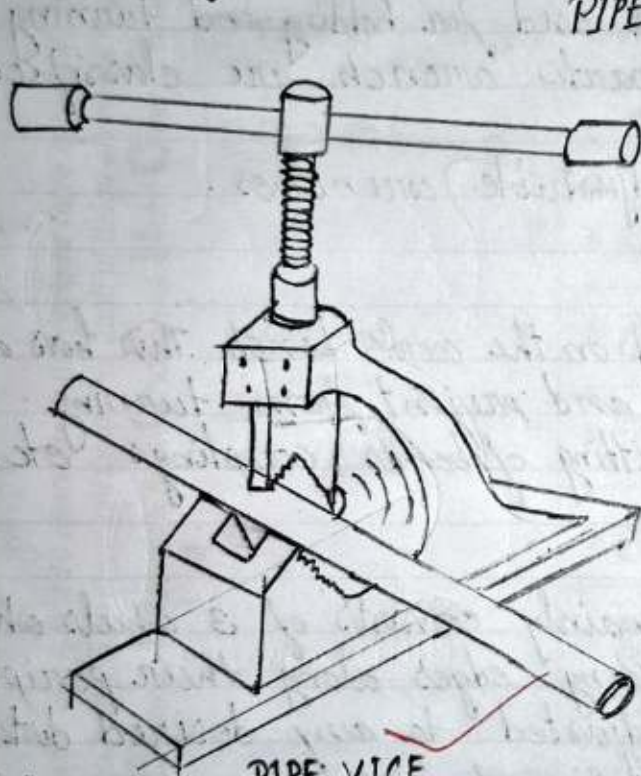


Fig: Pipe Cutter.



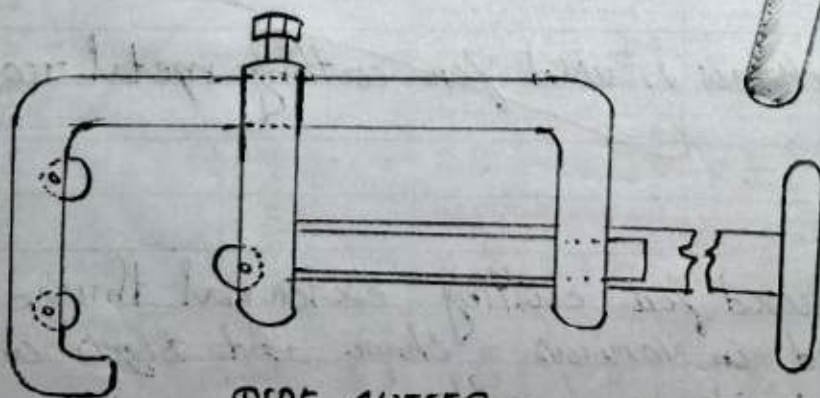
PIPE WRENCH



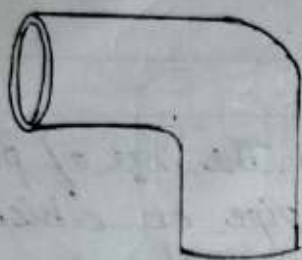
PIPE VICE



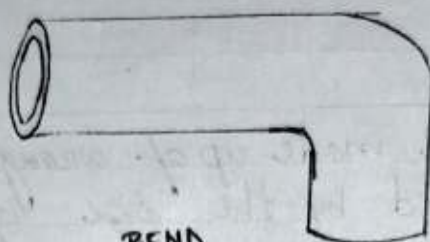
THREADING DIES



PIPE CUTTER



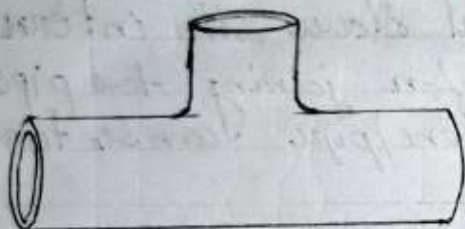
ELBOW



BEND



COUPLING

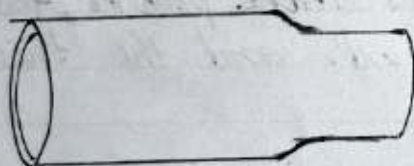


TEE



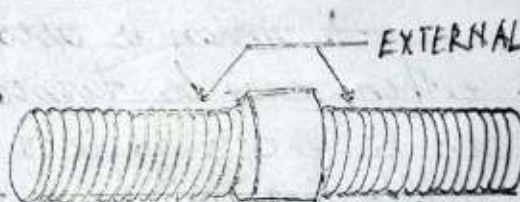
CAP

CLOSED END



REDUCER COUPLING

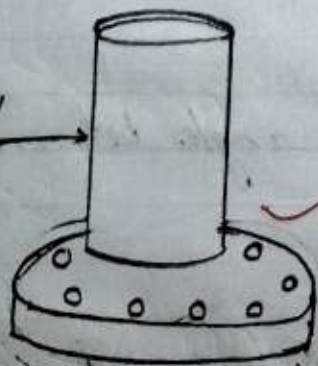
Good



NIPPLE

EXTERNAL V-THREADS

HUB WITH
THREER



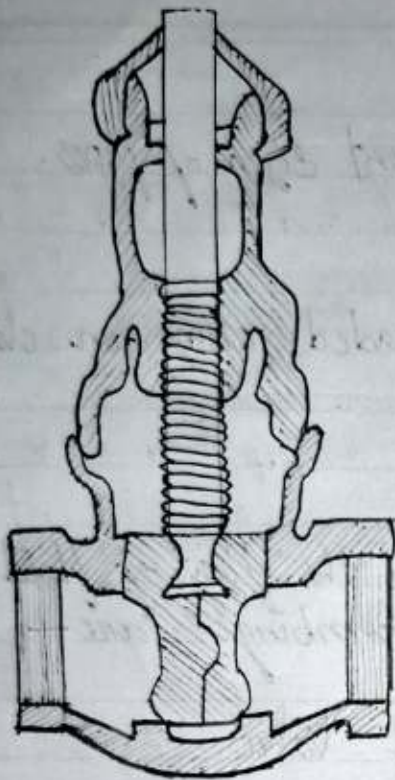
FLANGE



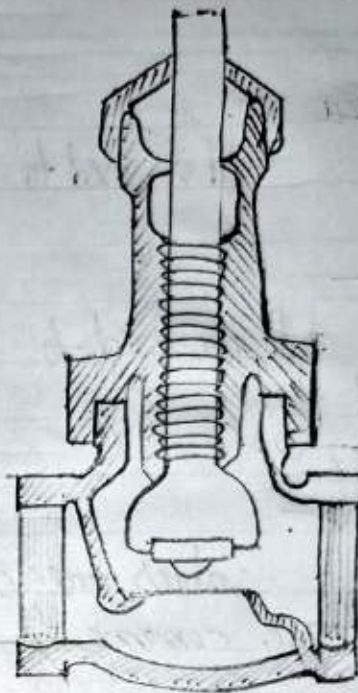
COUPLING

INTERNAL V-THREER

Disc with
HOLES

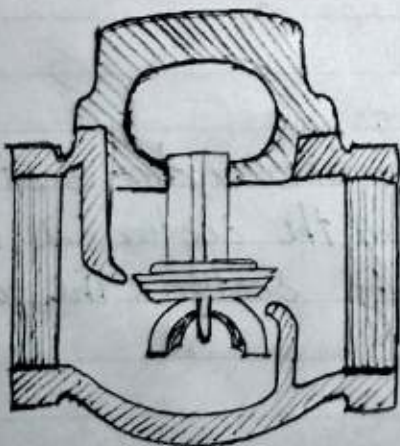


GATE VALVE

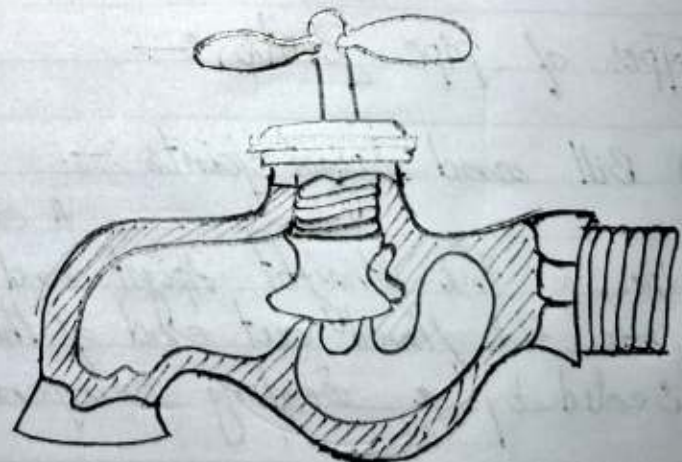


GLOBE VALVE

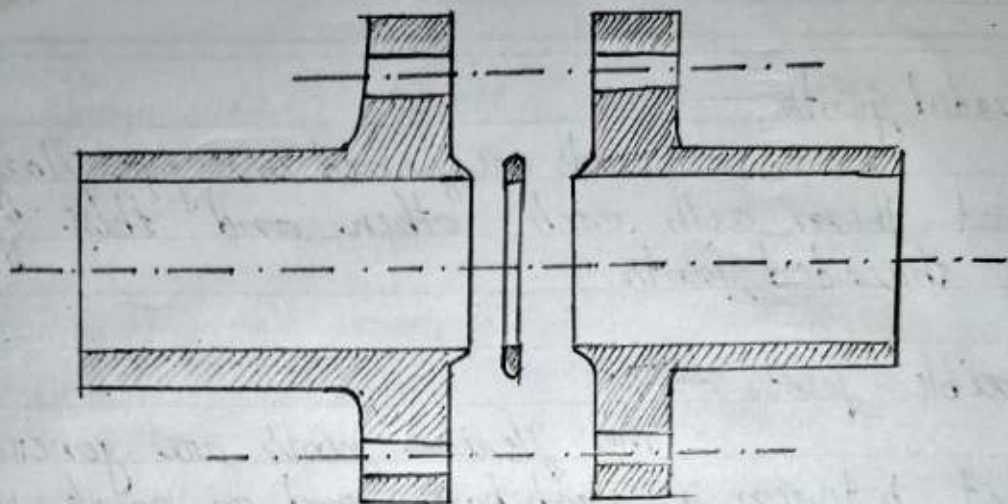
V. Good



CHECK VALVE

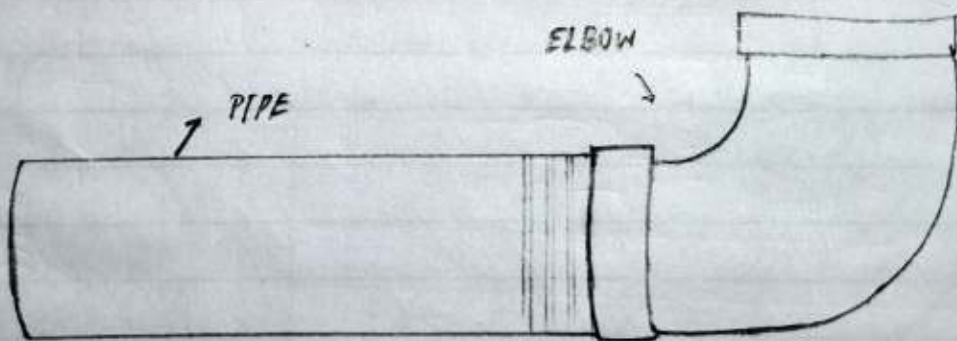


COMMON TAP



FLANGE JOINT

~~Copy~~



THREADED JOINT

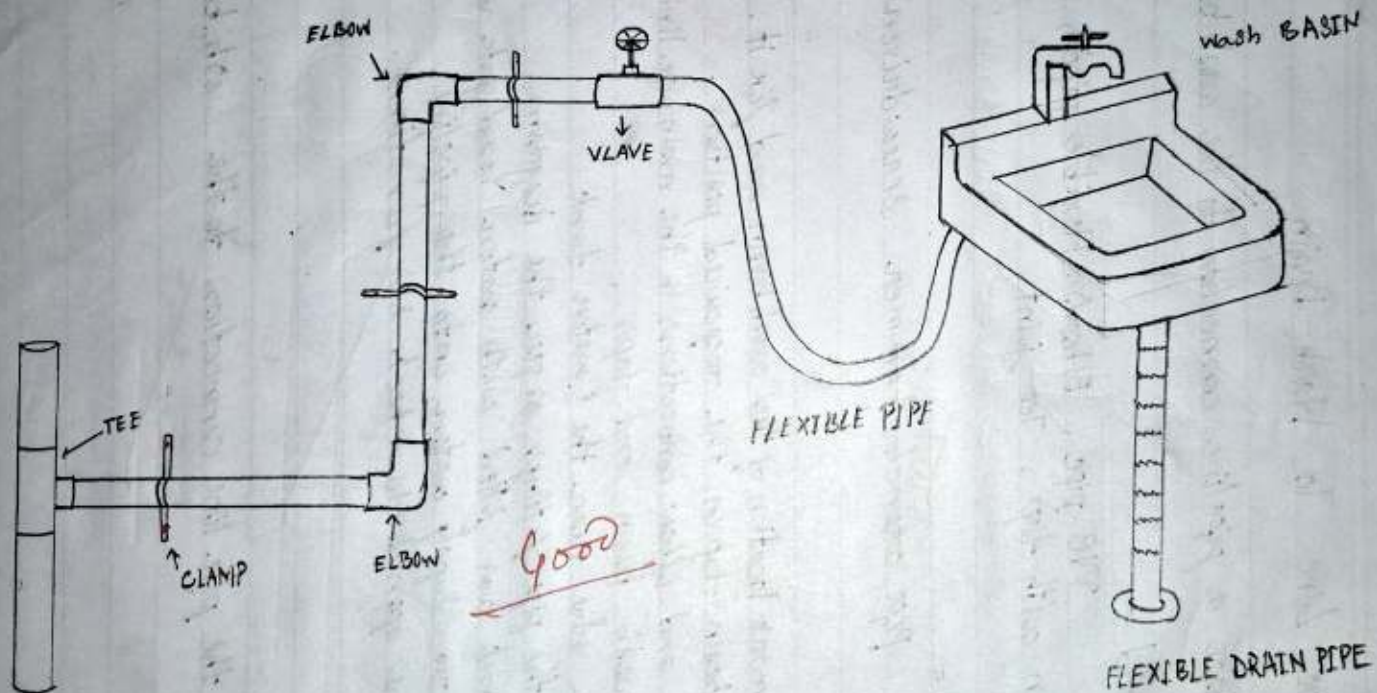
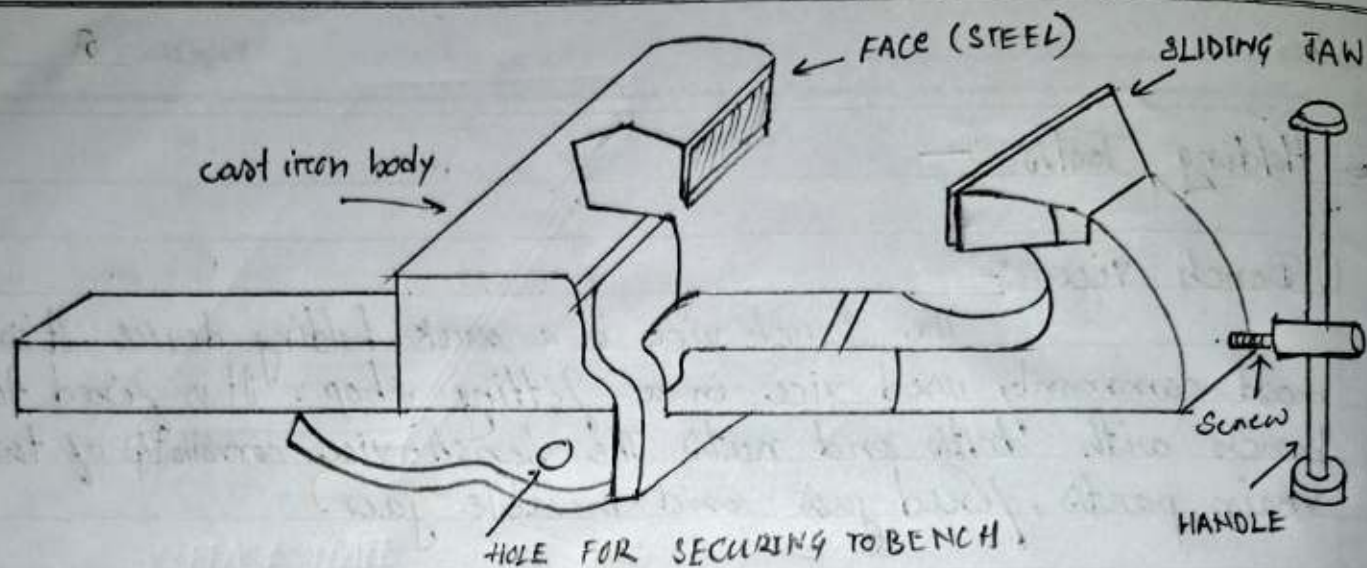
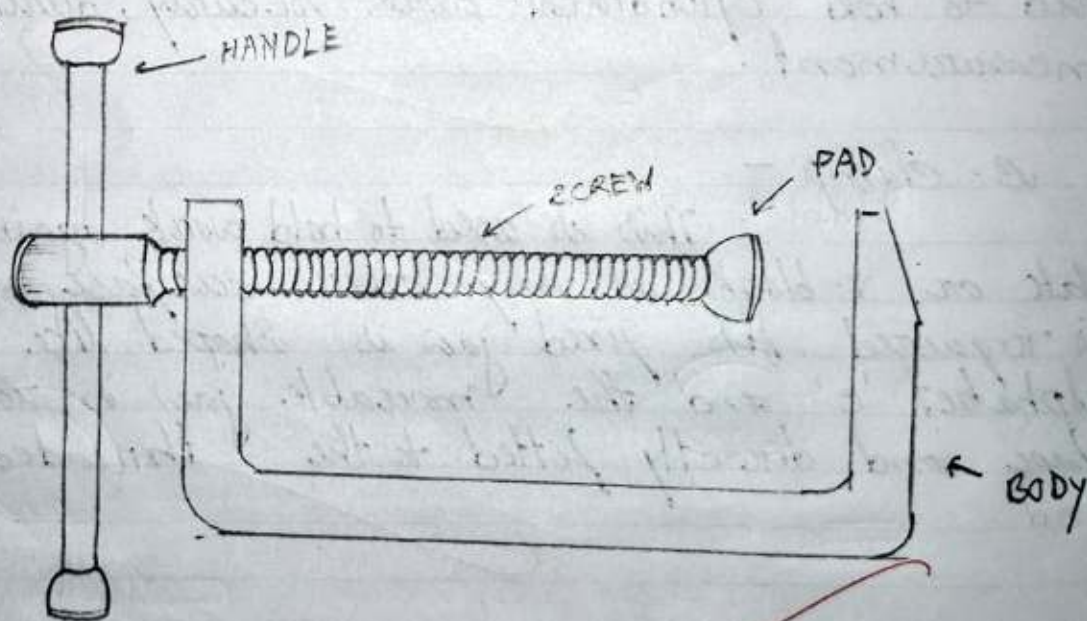


FIG : PIPELINE TO WASH BASIN

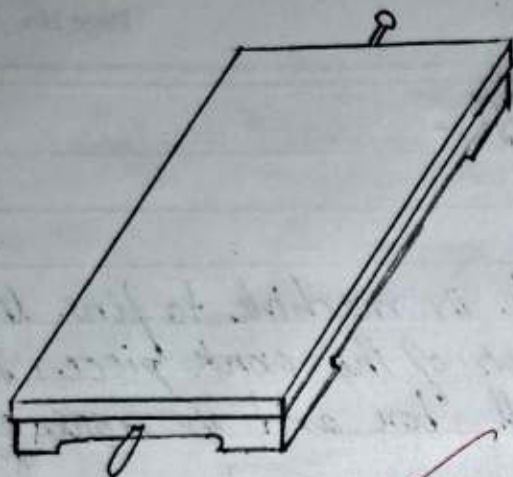


BENCH VICE

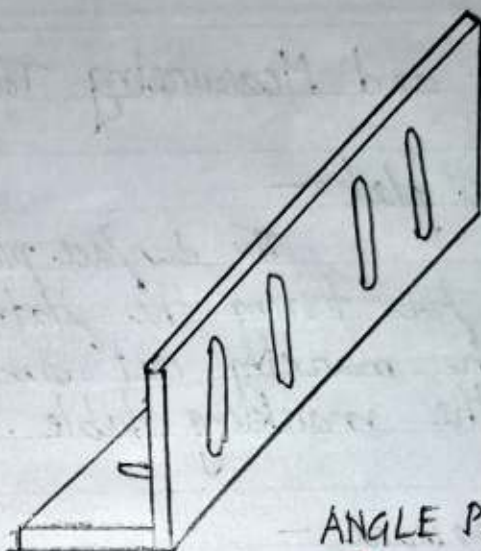
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C-CLAMP

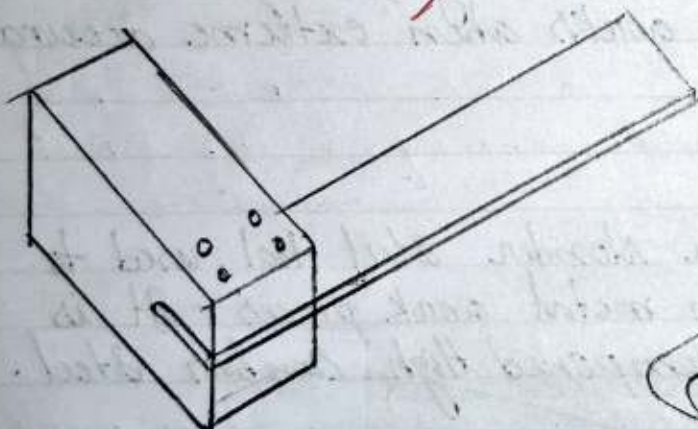


SURFACE PLATE

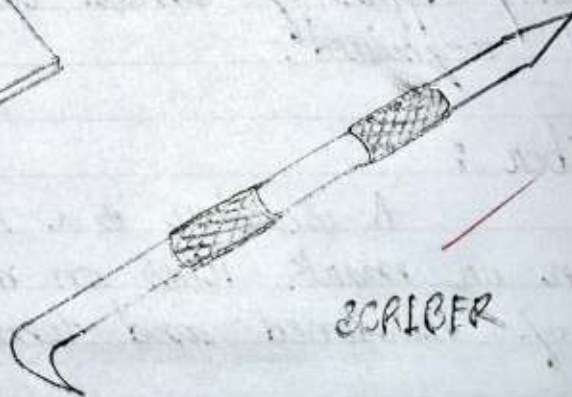


ANGLE PLATE

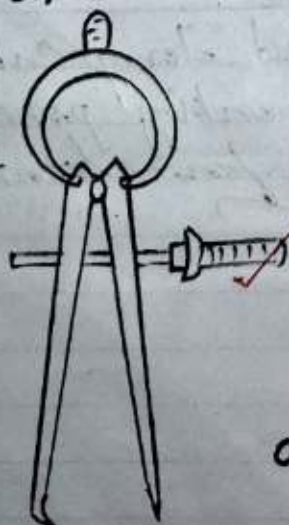
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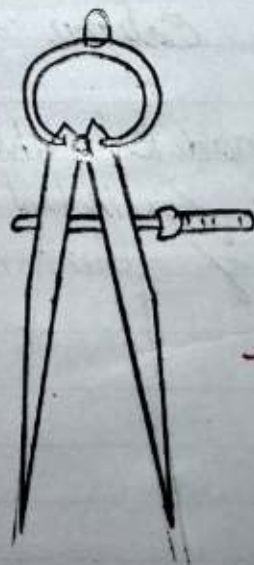
TRY SQUARE



SCRIBER



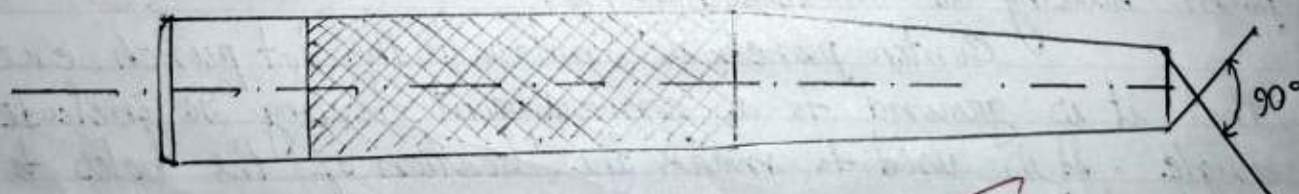
ODD LEG CALIPE AND
DIVIDER



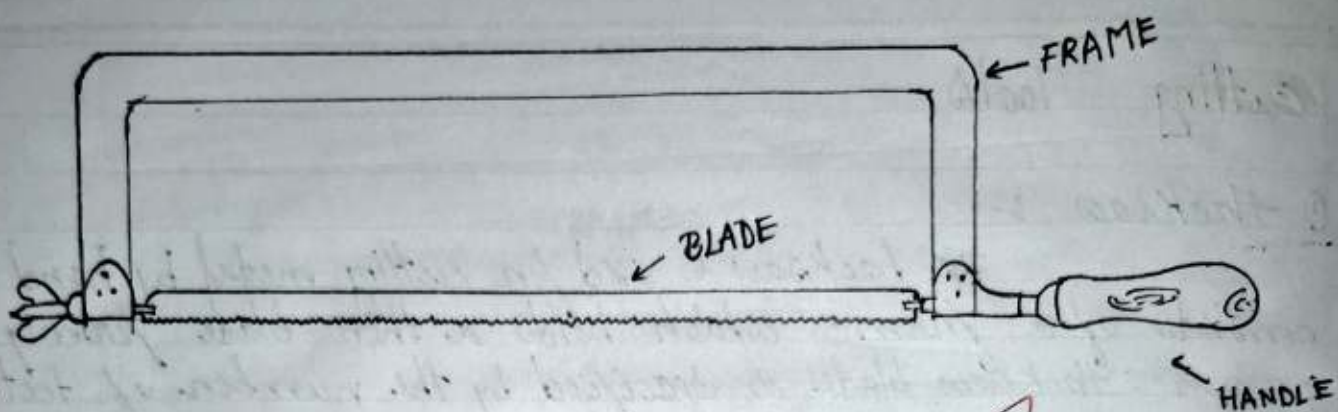


DOT PUNCH

Good



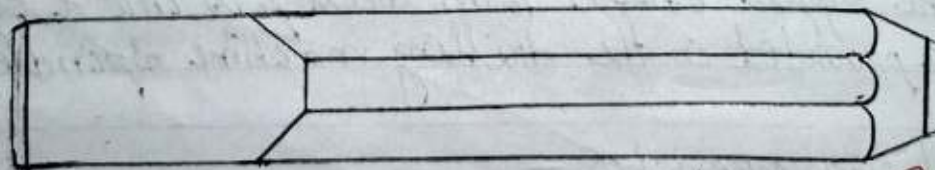
CENTER PUNCH.

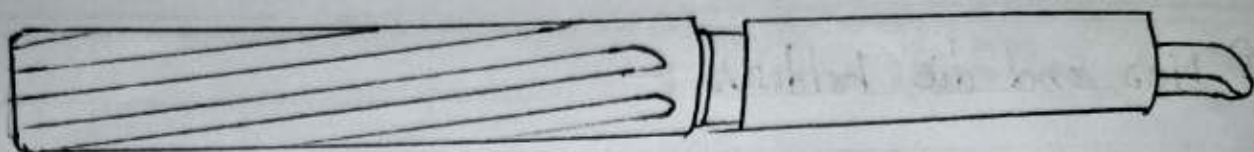


HACKSAE

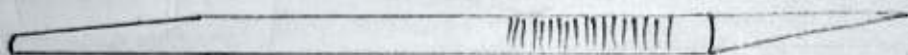
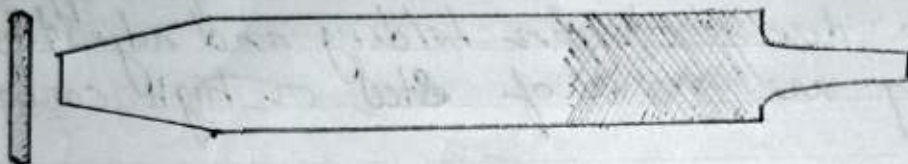


FLAT CHISEL

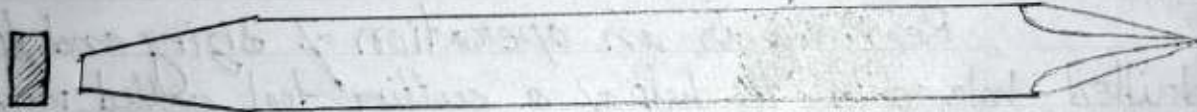




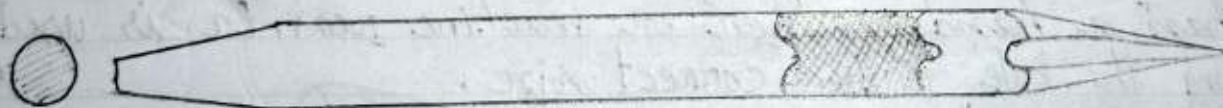
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FLAT FILE

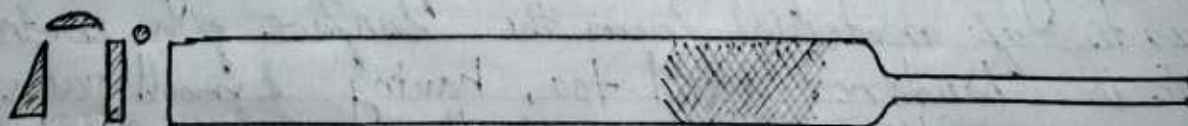


SQUARE FILE



ROUND FILE

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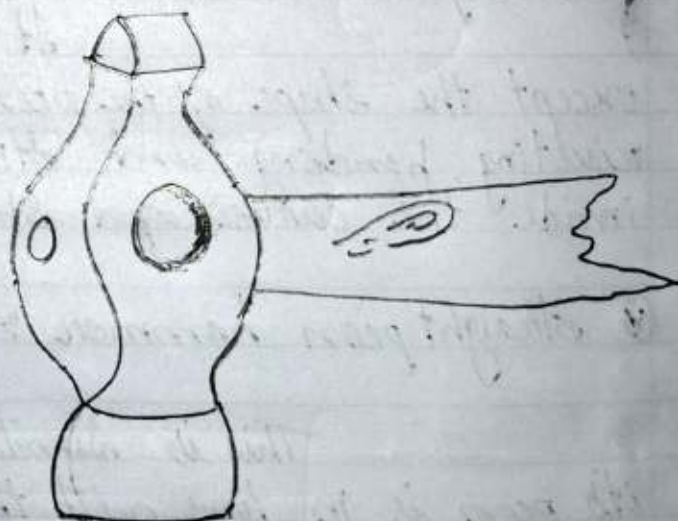
NEEDLE FILE



BALLPEEN HAMMER



CROSS PEEN HAMMER

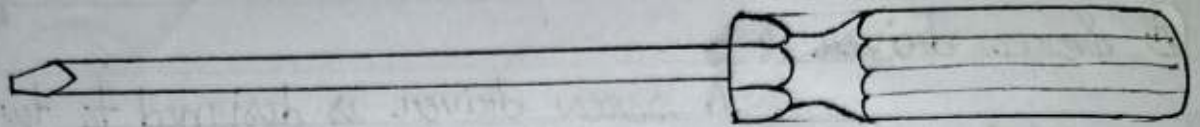


STRAIGHT PEEN HAMMER

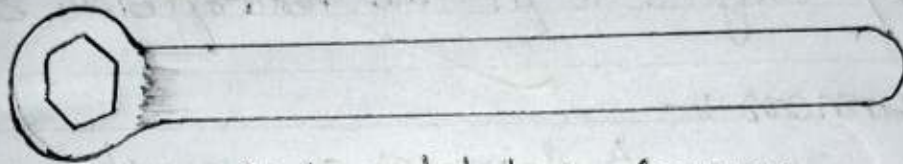
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Good

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SCREW DRIVER

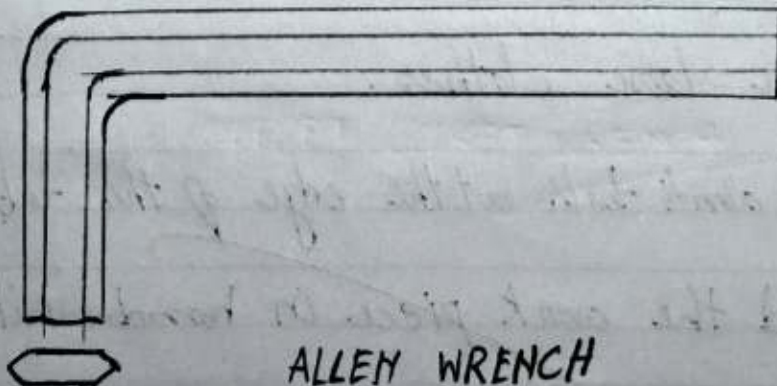


Single ended box spanner
HEXAGON

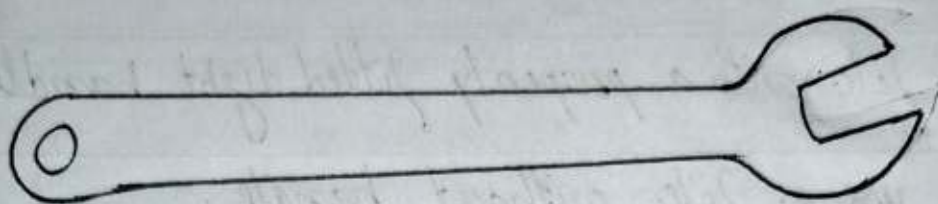
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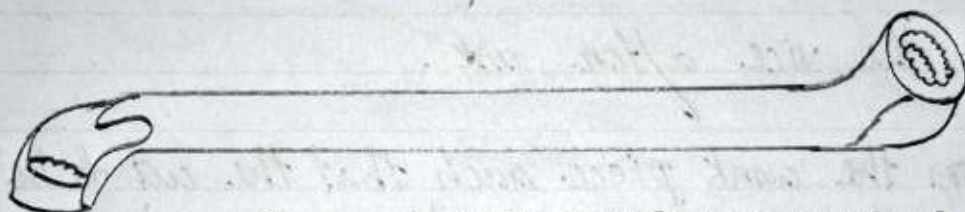
DOUBLE ENDED SPANNER



ALLEN WRENCH

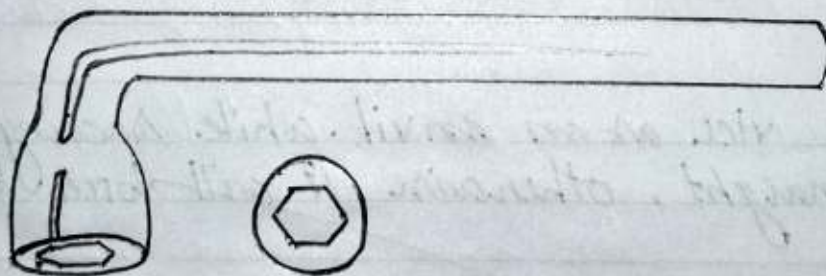


SINGLE ENDED SPANNER



OFFSET DOUBLE ENDED BOX SPANNER.

Good



OFFSET SOCKET WRENCH

Date - 8/9/25

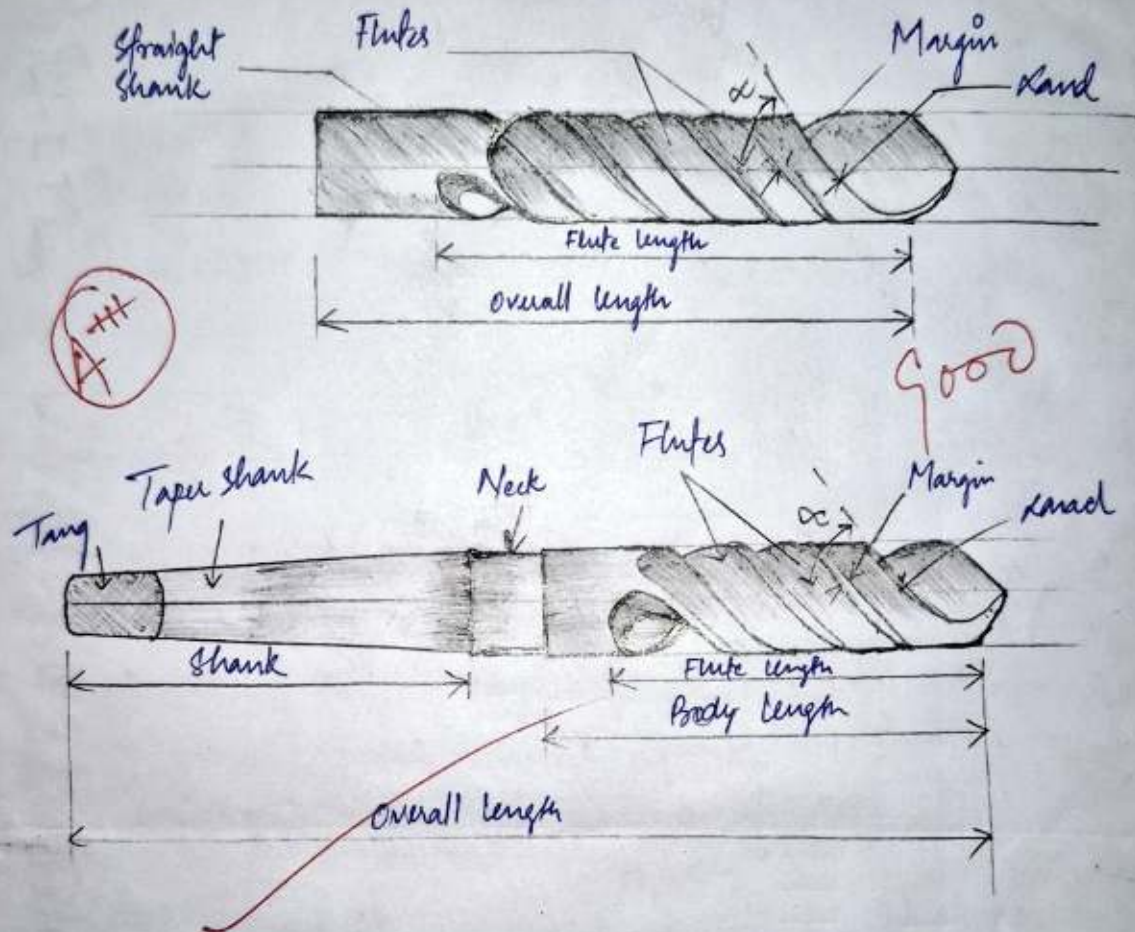


Figure : Twist Drill.

~~8/9/25~~